

Customer Shopping Behavior Analysis

Finding patterns in 3,900 purchases to inform strategic business choices

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OVERVIEW

Dataset at a Glance

3,900	18	$59.76	3.75
Total Purchases	**Data Points**	**Avg Purchase**	**Avg Rating**
Transactions analyzed	Features per purchase	Per transaction	Customer satisfaction

Comprehensive data spanning demographics, purchase details, and shopping behavior across all seasons

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PYTHON

Data Preparation & Cleaning

01

Data Loading

Imported dataset using pandas, explored structure with df.info() and summary statistics

02

Missing Data

Imputed 37 missing Review Ratings using median by product category

03

Standardization

Renamed columns to snake_case, dropped redundant promo_code_used field

04

Feature Engineering

Created age_group bins and purchase_frequency_days columns

05

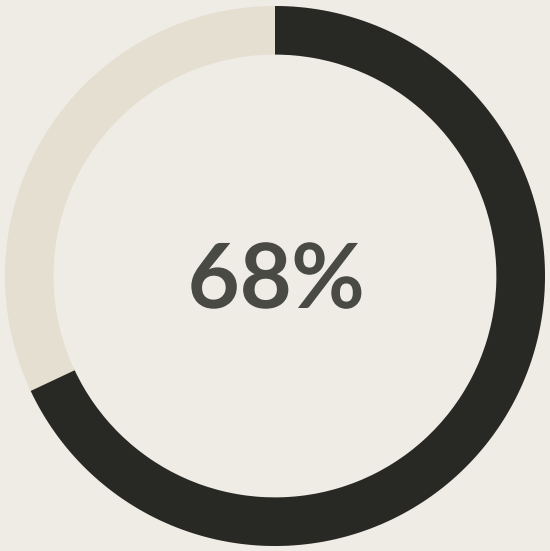
Database Integration

Connected to PostgreSQL and loaded cleaned data for SQL analysis

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SQL ANALYSIS

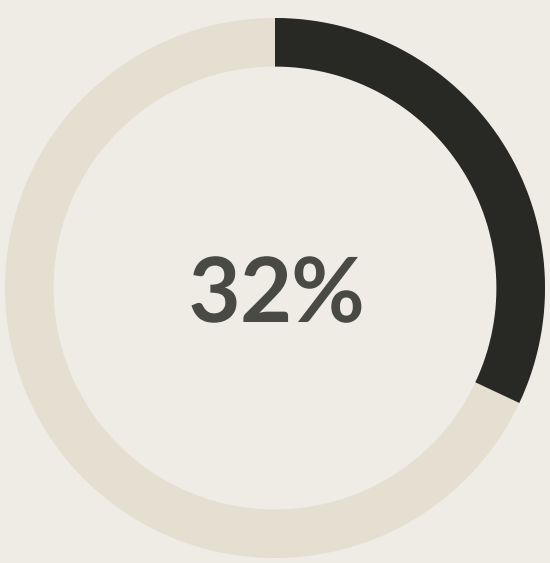
Revenue Insights



Male Revenue

$157,890 total





Female Revenue

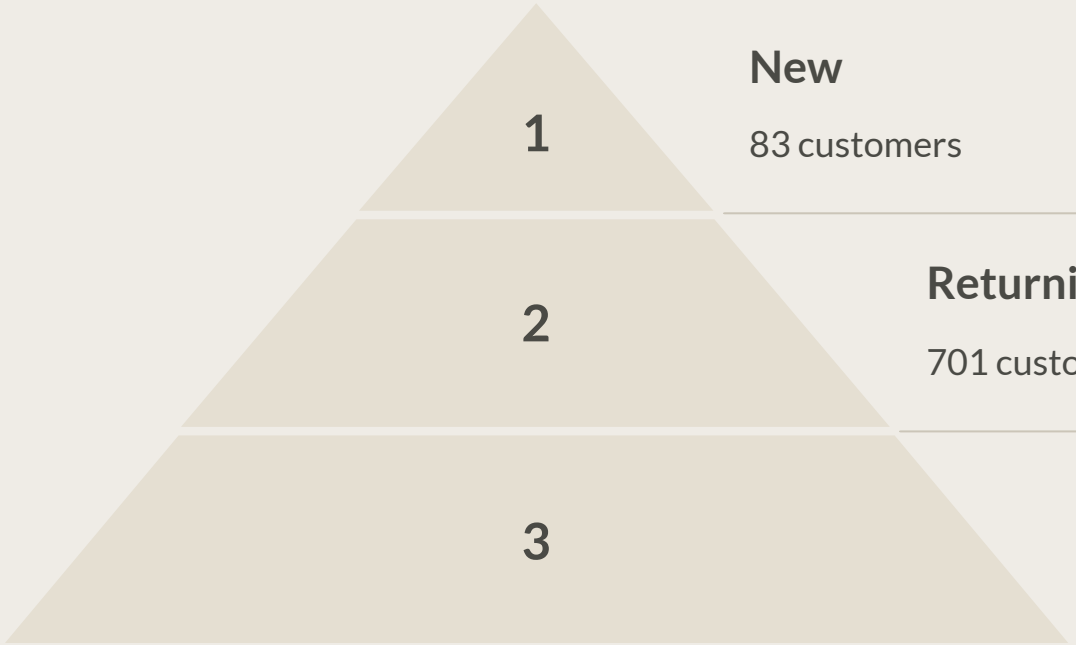
$75,191 total

Shipping Type Impact

- Express: $60.48 average
- Standard: $58.46 average

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Customer Segmentation



New

83 customers

Returning

701 customers

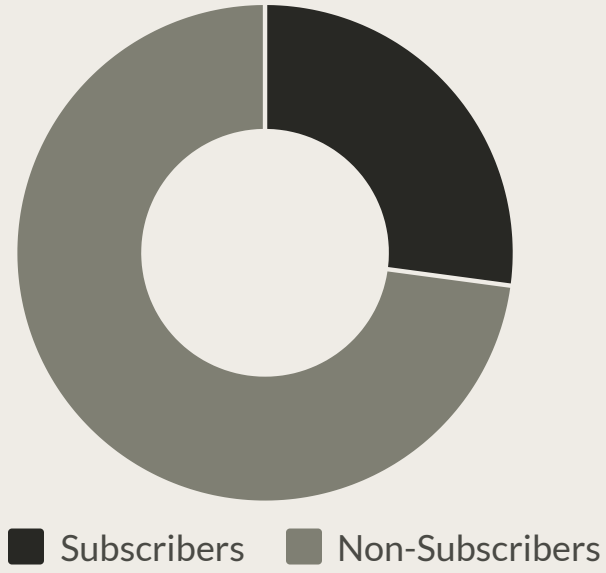
Loyal

3,116 customers

80% of customers classified as Loyal based on purchase history—strong retention foundation

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Subscription Analysis



Key Finding

Only 27% subscribe despite similar spending patterns

$59.49

Avg subscriber spend

$59.87

Avg non-subscriber spend

Opportunity: Convert high-spending non-subscribers

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TOP PERFORMERS

Product & Category Leaders

1

Highest Rated

Gloves (3.86), Sandals (3.84), Boots (3.82)

2

Top by Category

Clothing: Blouse (171 orders) | Accessories: Jewelry (171) | Footwear: Sandals (160)

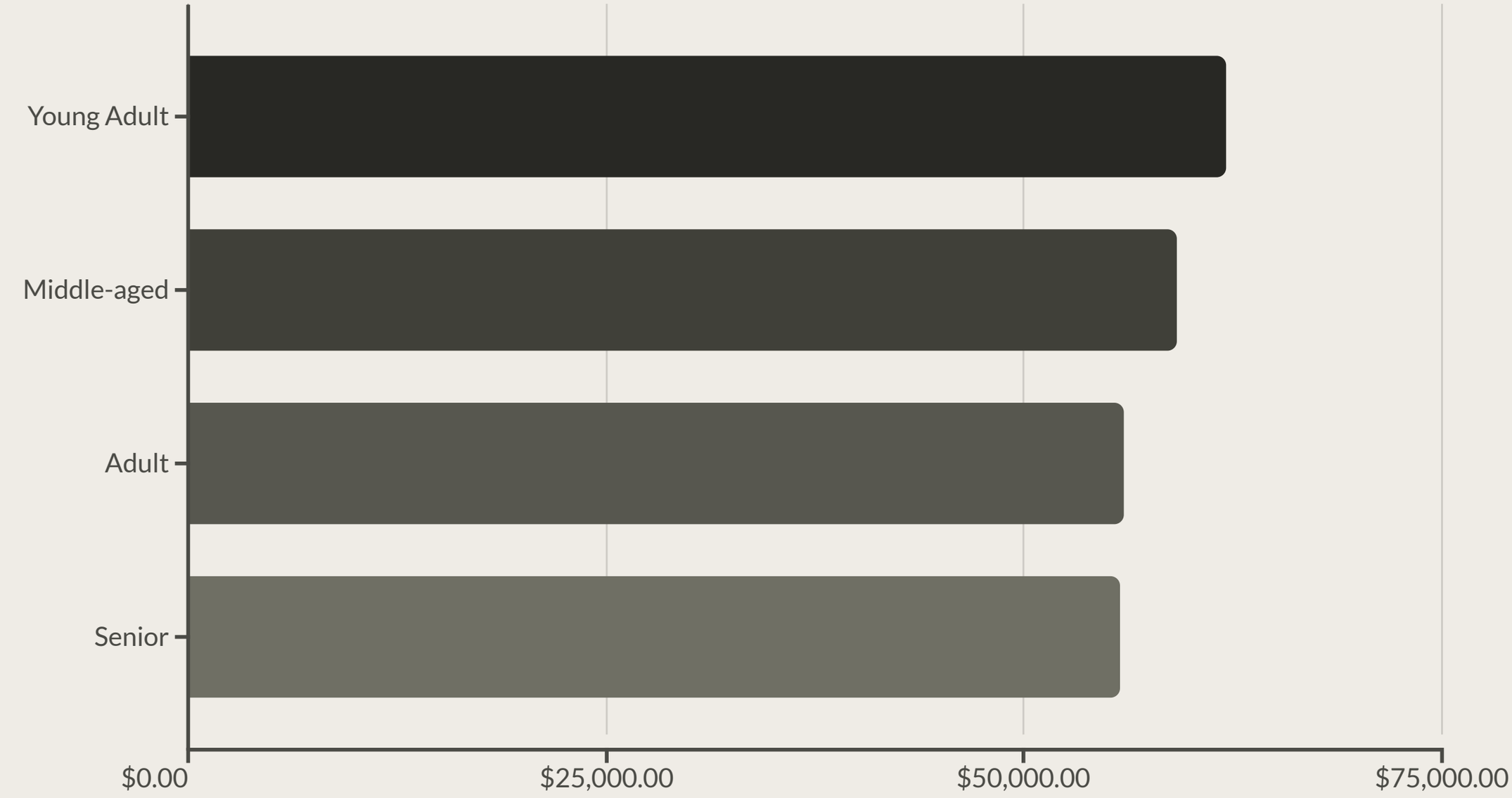
3

Discount Dependent

Hat (50%), Sneakers (49.66%), Coat (49.07%) rely heavily on promotions

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Revenue by Age Group



Young Adults lead revenue generation—prime target for marketing campaigns

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POWER BI

Interactive Dashboard

Dynamic Filters

Subscription, gender, category, shipping type

Revenue Analysis

By category, age group, real-time metrics

Customer Insights

Subscription status, behavior patterns

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RECOMMENDATIONS

Strategic Action Plan

Boost Subscriptions

Promote exclusive benefits—73% untapped potential

Loyalty Programs

Reward repeat buyers, move Returning to Loyal segment

Optimize Discounts

Balance promotions with margin control—839 high spenders use discounts

Targeted Marketing

Focus on Young Adults, express shipping users, top-rated products



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